Agile Development Process and Devops

FAT LAB

ISWE406L

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23MIS0339

Slot:L45+L48

SET 4

Q)Ensure wallet balance are update correctly after transaction.

CODE:

DOCKERFFILE:

```
FROM eclipse-temurin:17-jdk

COPY target/hello-world-crud-1.0.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

DEPLOYAMENT.YAML

```
apiVersion: apps/v1

kind: Deployment

metadata:

  name: hello-world-crud

spec:

  replicas: 1

  selector:

    matchLabels:

      app: hello-world-crud

  template:
```

```yaml
  metadata:
    labels:
      app: hello-world-crud
  spec:
    containers:
    - name: hello-world-crud
      image: sivaprakasamp06/hello-world-crud:latest
      imagePullPolicy: Always
```

POM.XML

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>


    <groupId>com.example</groupId>
    <artifactId>hello-world-crud</artifactId>
    <version>1.0</version>
```

```xml
    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>
```

```xml
    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.10.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
```

```xml
    <build>
        <plugins>
            <plugin>
```

```
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.2.5</version>
            </plugin>
            <plugin>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.3.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>Main</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

MAIN.JAVA

```
public class Main {
    public static void main(String[] args) throws InterruptedException {

        MessageManager manager = new MessageManager();
        manager.create(new Message(1, "wallet balance updated"));
```

```
        System.out.println("STATUS:FINISHED");
        System.out.println(manager.readAll().get(0).getText());
```

```
        // Keep container alive for Kubernetes
        while (true) {
            Thread.sleep(10000); // sleep 10 seconds
        }
    }
}
```

MESSAGE.JAVA

```java
public class Message {

    private int id;

    private String text;


    public Message(int id, String text) {

        this.id = id;

        this.text = text;

    }
```

```java
    public int getId() {

        return id;

    }
```

```java
    public String getText() {

        return text;

    }
```

```java
    public void setText(String text) {

        this.text = text;

    }

}
```

MESSAGEMANAGER.JAVA

```java
import java.util.*;


public class MessageManager {
```

```java
    private final Map<Integer, Message> messages = new HashMap<>();
```

```java
    public void create(Message message) {

        messages.put(message.getId(), message);

    }
```

```
    public List<Message> readAll() {

        return new ArrayList<>(messages.values());

    }
```

```
    public void update(int id, String newText) {

        if (messages.containsKey(id)) {

            messages.get(id).setText(newText);

        }

    }
```

```
    public void delete(int id) {

        messages.remove(id);

    }

}
```

MESSAGEMANAGERTEST.JAVA

```
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class MessageManagerTest {
```

```
    @Test
    void testHelloWorldMessage() {
        MessageManager manager = new MessageManager();
```

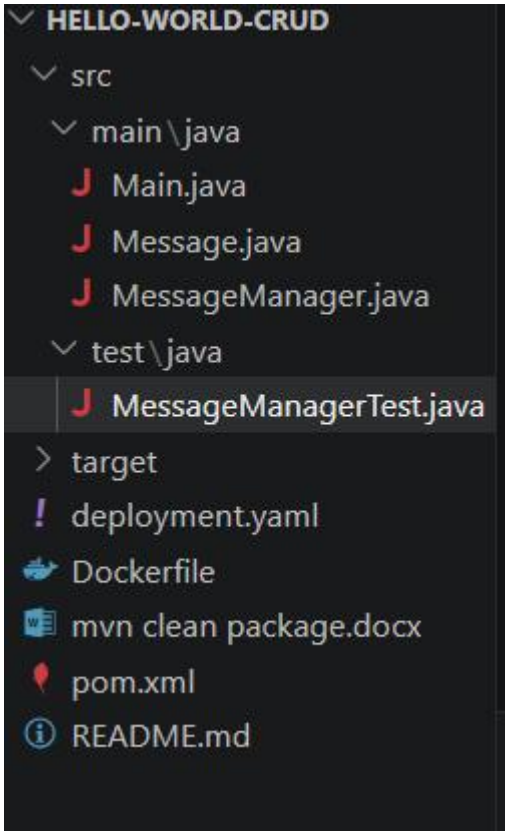
```
        manager.create(new Message(1, "wallet balance updated"));
```

```
        assertEquals("wallet balance updated", manager.readAll().get(0).getText());
    }
```

```
    @Test
    void testStatusFinished() {
        String status = "STATUS:FINISHED";
```

```
        assertEquals("STATUS:FINISHED", status);
    }
}
```

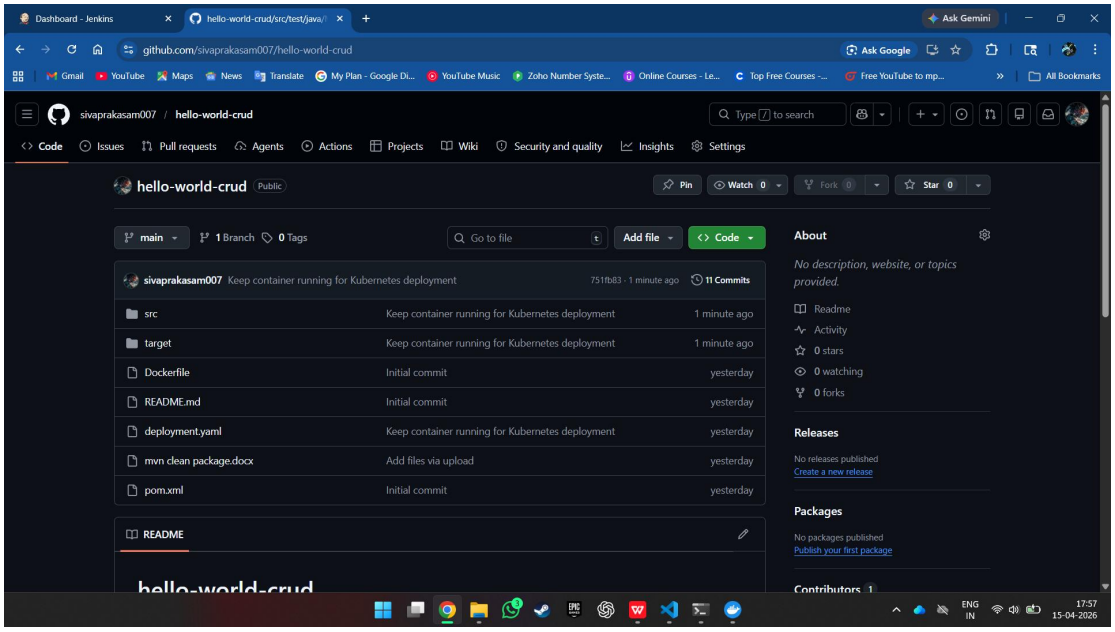
FILE STRUCTURE:



GIT

```
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> git init
>> git add .
>> git commit -m "Initial commit"
 create mode 100644 target/test-classes/MessageManagerTest.class
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> git remote add origin https://github.com/sivaprakasam007/hello-world-crud.git
>> git branch -M main
>> git push -u origin main
To https://github.com/sivaprakasam007/hello-world-crud.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/sivaprakasam007/hello-world-crud.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> git pull origin main --allow-unrelated-histories
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 868 bytes | 20.00 KiB/s, done.
From https://github.com/sivaprakasam007/hello-world-crud
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
```

```
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> git add .
>> git commit -m "Keep container running for Kubernetes deployment"
>> git push origin main
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time Git to
uches it
warning: in the working copy of 'target/surefire-reports/TEST-MessageManagerTest.xml', LF will be replaced by CRLF the next time Git touches it
[main 751fb83] Keep container running for Kubernetes deployment
 7 files changed, 9 insertions(+), 9 deletions(-)
Enumerating objects: 35, done.
Counting objects: 100% (35/35), done.
Delta compression using up to 8 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (18/18), 2.44 KiB | 833.00 KiB/s, done.
Total 18 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), completed with 8 local objects.
To https://github.com/sivaprakasam007/hello-world-crud.git
   ff9c50b..751fb83  main -> main
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud>
```



MAVEN:

```
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:hello-world-crud >-------------------
[INFO] Building hello-world-crud 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ hello-world-crud ---
[INFO] Deleting C:\Users\SIVA\OneDrive\Documents\hello-world-crud\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ hello-world-crud ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\SIVA\OneDrive\Documents\hello-world-crud\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ hello-world-crud ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 3 source files with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ hello-world-crud ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\SIVA\OneDrive\Documents\hello-world-crud\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ hello-world-crud ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ hello-world-crud ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
```

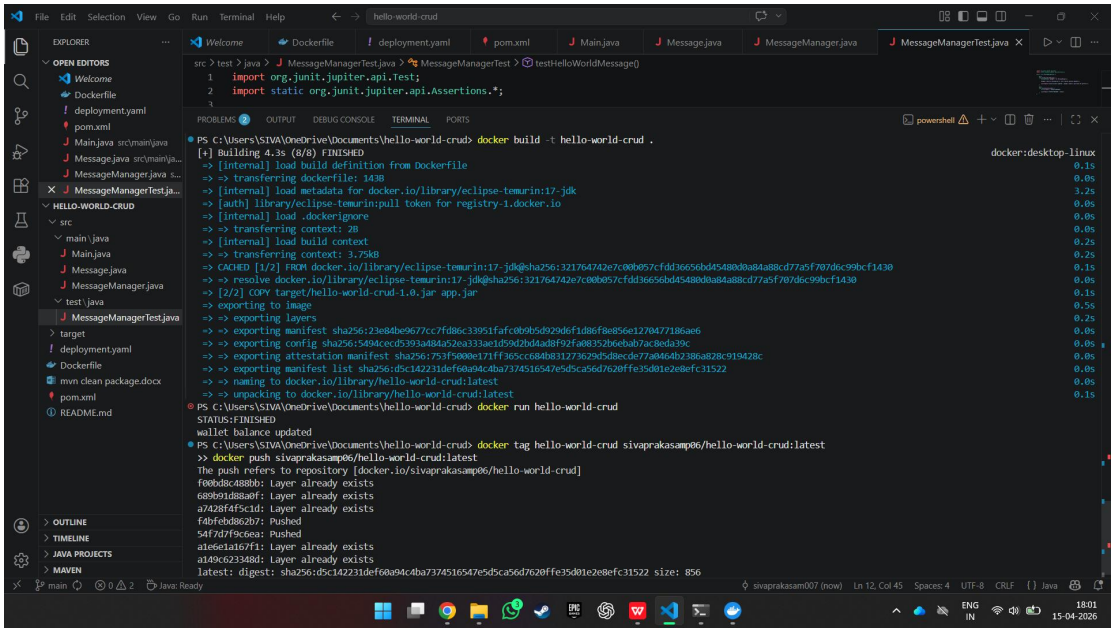
```
[INFO] Running MessageManagerTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.104 s -- in MessageManagerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ hello-world-crud ---
[INFO] Building jar: C:\Users\SIVA\OneDrive\Documents\hello-world-crud\target\hello-world-crud-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.175 s
[INFO] Finished at: 2026-04-15T17:53:38+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> java -jar target/hello-world-crud-1.0.jar
STATUS:FINISHED
wallet balance updated
```

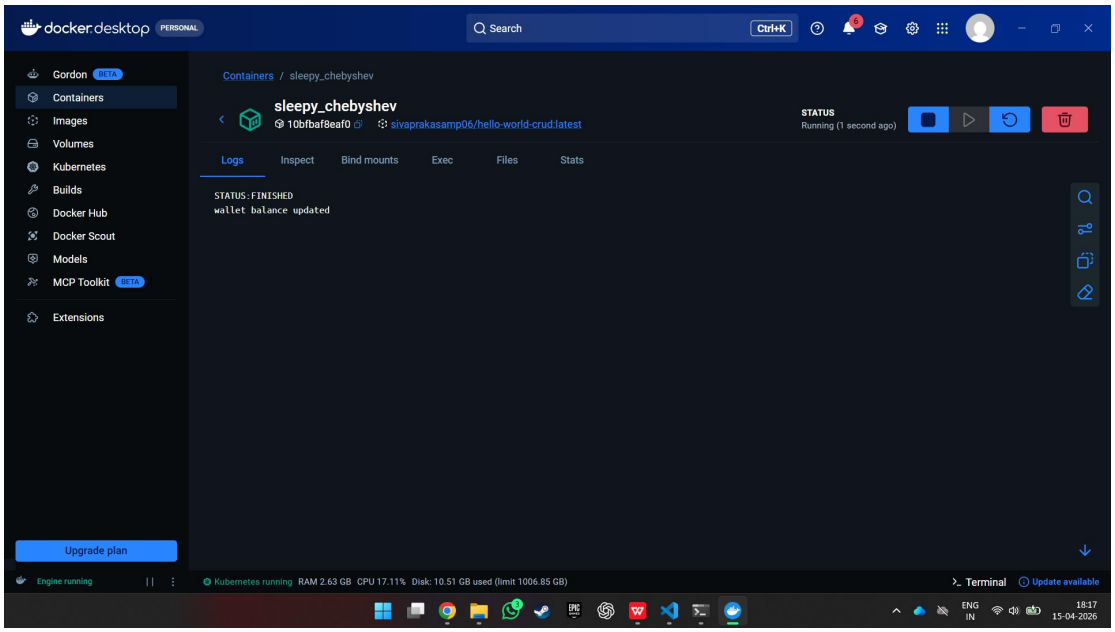
KUBERNETES:

```
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> docker tag hello-world-crud sivaprakasamp06/hello-world-crud:latest
>> docker push sivaprakasamp06/hello-world-crud:latest
The push refers to repository [docker.io/sivaprakasamp06/hello-world-crud]
f00bd8c488bb: Layer already exists
689b91d88a0f: Layer already exists
a7428f4f5c1d: Layer already exists
f4bfebd862b7: Pushed
54f7d7f9c6ea: Pushed
a1e6e1a167f1: Layer already exists
a149c623348d: Layer already exists
latest: digest: sha256:d5c142231def60a94c4ba7374516547e5d5ca56d7620ffe35d01e2e8efc31522 size: 856
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud created
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
hello-world-crud-775fb9dbbf-9jwln   0/1     ContainerCreating   0          11s
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud unchanged
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl rollout restart deployment hello-world-crud
deployment.apps/hello-world-crud restarted
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud unchanged
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-world-crud-855654c54c-dx7vz   1/1     Running   0          7s
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl logs ^C
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl logs hello-world-crud-855654c54c-dx7vz
STATUS:FINISHED
wallet balance updated
```

```
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-world-crud-855654c54c-dx7vz   1/1     Running   0          7s
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl logs ^C
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> kubectl logs hello-world-crud-855654c54c-dx7vz
STATUS:FINISHED
wallet balance updated
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud>
```

DOCKER:


PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> docker build -t hello-world-crud .
[+] Building 4.3s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> [internal] load build context
=> [2/2] COPY target/hello-world-crud-1.0.jar app.jar
=> exporting to image
=> exporting layers
=> exporting manifest sha256:23e84be9677cc7fd86c33951fafc0b9b5d929d6f1d86f8e856e1270477186ae6
=> exporting config sha256:5494ced5393a484a52ea333ae1d59d2bd4ad8f92fa08352b6ebab7ac8eda39c
=> exporting attestation manifest sha256:753f5000e171ff365cc684b831273629d5d8ecde77a0464b2386a828c919428c
=> exporting manifest list sha256:d5c142231def60a94c4ba7374516547e5d5ca56d7620ffe35d01e2e8efc31522
=> naming to docker.io/library/hello-world-crud:latest
=> unpacking to docker.io/library/hello-world-crud:latest
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> docker run hello-world-crud
STATUS:FINISHED
wallet balance updated
PS C:\Users\SIVA\OneDrive\Documents\hello-world-crud> docker tag hello-world-crud sivaprakasamp06/hello-world-crud:latest
>> docker push sivaprakasamp06/hello-world-crud:latest
The push refers to repository [docker.io/sivaprakasamp06/hello-world-crud]
f00bd8c488bb: Layer already exists
689b91d88a0f: Layer already exists
a7428f4f5c1d: Layer already exists
f4bfebd862b7: Pushed
54f7d7f9c6ea: Pushed
a1e6e1a167f1: Layer already exists
a149c623348d: Layer already exists
latest: digest: sha256:d5c142231def60a94c4ba7374516547e5d5ca56d7620ffe35d01e2e8efc31522 size: 856


docker desktop
Containers / sleepy_chebyshev
sleepy_chebyshev
10bfbaf8eaf0
sivaprakasamp06/hello-world-crud:latest
STATUS
Running (1 second ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
STATUS:FINISHED
wallet balance updated

kindest/node v1.34.3 08497ee19eac 4 months ago 1.35 GB
docker/desktop-cloud-provider-kind v0.5.0 4ad59ce20658 4 months ago 595.06 MB
envoyproxy/envoy v1.36.4 08acc53db6f4 4 months ago 247.74 MB
docker/desktop-containerd-registry v0.0.3 179a9ea3ae47 4 months ago 50.32 MB
hello-world-crud latest d5c142231def 17 minutes ag 635.92 MB
sivaprakasamp06/hello-world-crud latest 6b5bfd51e6d0 6 minutes ago 635.92 MB
Showing 13 items

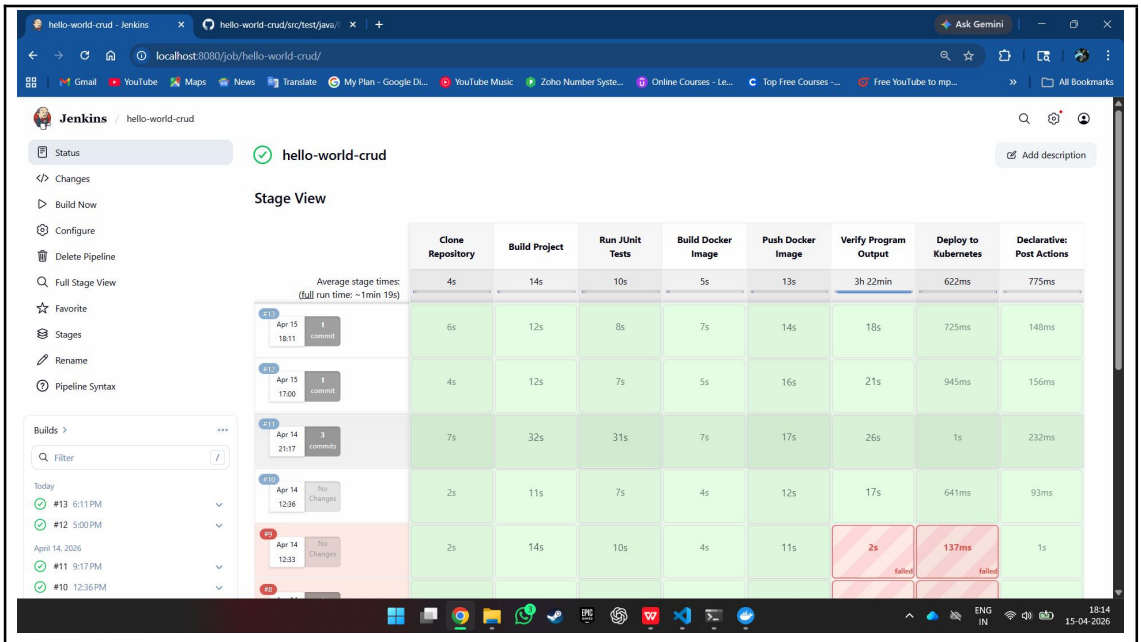
JENKINES:

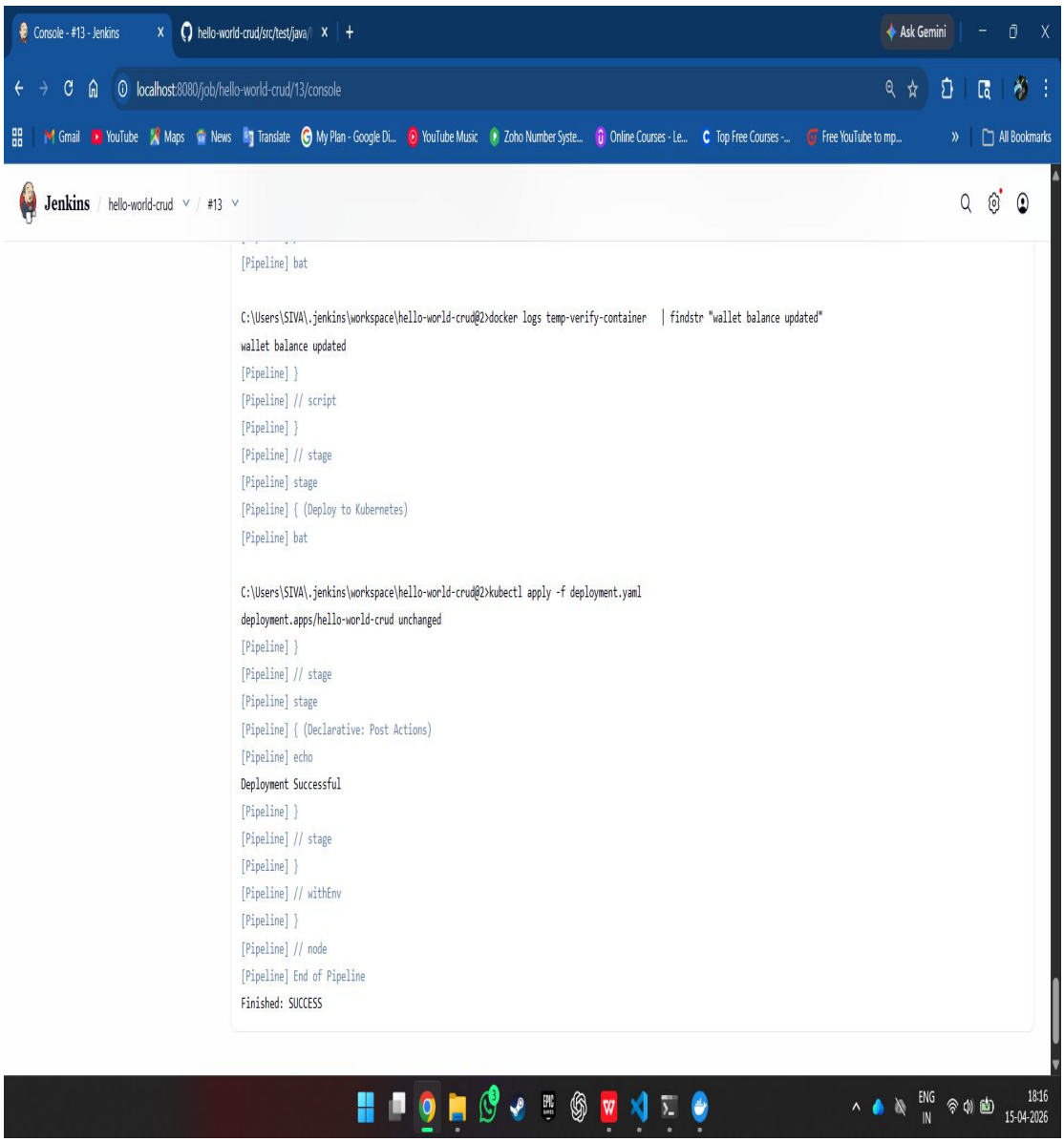
Pipeline script:

```
pipeline {
   agent any


   environment {
      DOCKER_IMAGE = 'sivaprakasamp06/hello-world-crud:latest'
   }


   stages {
      stage('Clone Repository') {
         steps {
            git branch: 'main',
               url: 'https://github.com/sivaprakasam007/hello-world-crud.git'
         }
      }


      stage('Build Project') {
         steps {
            bat 'mvn clean package'
         }
      }


      stage('Run JUnit Tests') {
         steps {
            bat 'mvn test'
         }
      }
```

```
    stage('Build Docker Image') {
        steps {
            bat 'docker build -t %DOCKER_IMAGE% .'
        }
    }

    stage('Push Docker Image') {
        steps {
            withCredentials([usernamePassword(
                credentialsId: 'dockerhub-creds',
                usernameVariable: 'DOCKER_USER',
                passwordVariable: 'DOCKER_PASS'
            )]) {
                retry(3) {
                    bat 'docker login -u %DOCKER_USER% -p %DOCKER_PASS%'
                    bat 'docker push %DOCKER_IMAGE%'
                }
            }
        }
    }

   stage('Verify Program Output') {
  steps {
    script {
        bat 'docker rm -f temp-verify-container || ver >nul'
        bat 'docker run -d --name temp-verify-container sivaprakasamp06/hello-
world-crud:latest'
```

```
        powershell 'Start-Sleep -Seconds 10'

        bat 'docker logs temp-verify-container | findstr "wallet balance updated"'
      }
    }
}

      stage('Deploy to Kubernetes') {
        steps {
          bat 'kubectl apply -f deployment.yaml'
        }
      }
    }

    post {
      success {
        echo 'Deployment Successful'
      }
      failure {
        // Ensure cleanup happens even if the findstr check fails
        bat 'docker stop temp-verify-container || ver > nul'
        bat 'docker rm temp-verify-container || ver > nul'
        echo 'Pipeline Failed'
      }
    }
}
```


hello-world-crud - Jenkins
hello-world-crud/src/test/java/
Ask Gemini
localhost:8080/job/hello-world-crud/
Jenkins / hello-world-crud
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Favorite
Stages
Rename
Pipeline Syntax
Builds
Filter
Today
#13 6:11 PM
#12 5:00 PM
April 14, 2026
#11 9:17 PM
#10 12:36 PM
hello-world-crud
Add description
Stage View
Clone Repository
Build Project
Run JUnit Tests
Build Docker Image
Push Docker Image
Verify Program Output
Deploy to Kubernetes
Declarative: Post Actions
Average stage times:
(full run time: ~1min 19s)
4s
14s
10s
5s
13s
3h 22min
622ms
775ms
#13 Apr 15 18:11 1 commit 6s 12s 8s 7s 14s 18s 725ms 148ms
#12 Apr 15 17:00 1 commit 4s 12s 7s 5s 16s 21s 945ms 156ms
#11 Apr 14 21:17 3 commits 7s 32s 31s 7s 17s 26s 1s 232ms
#10 Apr 14 12:36 No Changes 2s 11s 7s 4s 12s 17s 641ms 93ms
#9 Apr 14 12:33 No Changes 2s 14s 10s 4s 11s 2s failed 137ms failed 1s
18:14
15-04-2026


Console - #13 - Jenkins
hello-world-crud/src/test/java/
Ask Gemini
localhost:8080/job/hello-world-crud/13/console
Jenkins / hello-world-crud / #13
[Pipeline] bat
C:\Users\SIVA\.jenkins\workspace\hello-world-crud@2>docker logs temp-verify-container | findstr "wallet balance updated"
wallet balance updated
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat
C:\Users\SIVA\.jenkins\workspace\hello-world-crud@2>kubectl apply -f deployment.yaml
deployment.apps/hello-world-crud unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment Successful
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
18:16
15-04-2026